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LARSON—MATH 310—CLASSROOM WORKSHEET 08
Linear Dependence, Linear Independence and Basis.

Review

1. What is the Cauchy-Schwarz Inequality?
2. What is the angle $\angle(\vec{a}, \vec{b})$ between vectors $\vec{a}$ and $\vec{b}$?
3. What is the definition of the *linear dependence* of vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_k$?

New (Chp. 5)

1. Check that vectors $\vec{a}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\vec{a}_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are linearly dependent.
2. Check that one of $\vec{a}_1, \vec{a}_2, \vec{a}_3$ can be written in terms of the others.
3. Argue: if vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_k$ are linearly dependent then one can be written as a linear combination of the others.
4. What is the definition of *linearly independent* vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_k$?
5. Check that $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are linearly independent.

6. Check that $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are linearly independent.

7. Check that $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\vec{a} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ are linearly independent.

8. Check that $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\vec{a} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ are linearly **dependent**. (What's the difference from the last example?)

9. We showed that vectors $\vec{a}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ are linearly independent.

- Let $\vec{x} = \vec{a}_1 + 2\vec{a}_2$. Find $\vec{x}$.
- Suppose $\vec{x} = \beta_1\vec{a}_1 + \beta_2\vec{a}_2$. Find β_1, β_2 .

10. Show: if $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_k$ are linearly independent and $\vec{x}$ is a linear combination of them (so $\vec{x} = \beta_1\vec{a}_1 + \dots + \beta_k\vec{a}_k$) then the coefficients $\beta_1, \dots, \beta_k$ are unique.

11. What is a *basis*?